



Certificado de Qualidade Quality Certificate



769022/2020

Cliente (Customer): MACOM INSTRUMENTAL CIRURGICO IND LTDA

Nota Fiscal (Invoice): 380403

Pedido Cliente (Customer order): 12.969

Pedido(Ordem): 748760

Data (Date): 11/02/2020

Material (Product)

Descricao (Description): Chapa 420 BF 1,50 mm x 1240 mm x 2000 mm (BN 19/01914/1)-(10 pcs) Num. Rast. BN 19/01914/1
Corrida (Heat): 970914H Placa/O.P.: 2003B Norma (Standard): ASTM A240/A240M Peso (Weight): 299,00 KG
Observacao (Observation):

Análise Química (Chemical Composition)

C	Si	Mn	Cr	Ni	Mo	N	Ti	P
0,313	0,370	0,32	12,07	0,15	0,012	112PPA	0,0029	0,0251
S	Cu	Al	N2	Nb	CO			
0,0032	0,0289				0,015			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2%
Resistencia a Tracao (Strizione): Dureza: 85,33 HRB
Reduc. Area Minima (Reduction of Area): Estricção
Alongamento (Elongation): Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 420 - 6,35 mm (1/4") (BR 12/00263/1)-(24 pcs) Num. Rast. BR 12/00263/1
Corrida (Heat): 25056 Placa/O.P.: Norma (Standard): ASTM A276-10 Peso (Weight): 15,00 KG
Observacao (Observation):

Análise Química (Chemical Composition)

C	Si	Mn	Cr	Ni	Mo	N	Ti	P
0,150	0,460	0,75	12,14	0,29	0,03			0,021
S	Cu	Al	N2	Nb	CO			
0,01	0,08		0,0040C		0,05			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 631 MPa
Resistencia a Tracao (Strizione): 718 MPa Dureza: 213 HB
Reduc. Area Minima (Reduction of Area): 65,0% Estricção
Alongamento (Elongation): 25,0% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 7,94 mm (5/16") (BR 19/02947/1)-(10 pcs) Num. Rast. BR 19/02947/1
Corrida (Heat): 112704 Placa/O.P.: Norma (Standard): Peso (Weight): 11,70 KG
Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO A 1050°C

Análise Química (Chemical Composition)

C	Si	Mn	Cr	Ni	Mo	N	Ti	P
0,023	0,530	1,53	18,15	8,05	0,37	0,0720		0,037
S	Cu	Al	N2	Nb	Co			
0,024	0,7				0,15			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 650,00
Resistencia a Tracao (Strizione): 772,00 Dureza: 228 HBW
Reduc. Area Minima (Reduction of Area): 71,00 Estricção
Alongamento (Elongation): 37,00 Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 9,53 mm (3/8") (BR 19/02584/1)-(24 pcs) Num. Rast. BR 19/02584/1
Corrida (Heat): 0812951 Placa/O.P.: Norma (Standard): ASTM A276/17 Peso (Weight): 33,60 KG
Observacao (Observation): BARRA REDONDA / LAMINADOS / SOLUBILIZADO NA LAMINAÇÃO / TREFILADO RETIFICADO

Análise Química (Chemical Composition)

C	Si	Mn	Cr	Ni	Mo	N	Ti	P
0,017	0,230	1,49	18,07	8,36	0,45	0,091		0,03
S	Cu	Al	N2	Nb	Co	V	W	
0,027	0,65				0,23	0,07	0,1	

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 692 MPa
Resistencia a Tracao (Strizione): 840 MPa Dureza: 276,5 HBW
Reduc. Area Minima (Reduction of Area): Estricção 71,3%
Alongamento (Elongation): 31,9% Tam. do Grao (Grain Size):





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Nota Fiscal (Invoice): 380403

Pedido Cliente (Customer order): 12.969

Pedido(Order): 748760

Data (Date): 11/02/2020

Material (Product)

Descricao (Description): Barra Redonda 304L - 28,57 mm (1.1/8") (BR 19/02060/1)-(10 pcs)

Num. Rast. BR 19/02060/1

Corrida (Heat): 53335

Placa/O.P.:

Norma (Standard): ASTM A276/M-17 A

Peso (Weight): 155,00 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Analise Quimica (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,016	0,350	1,55	18,2	8,08	0,34	0,0700		0,037
S	Cu	Al	N2	Nb	Co			
0,022	0,57				0,16			

Analise Mecanica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 494 MPa
Resistencia a Tracao (Strizione): 675 MPa Dureza: 200 HBW
Reduc. Area Minima (Reduction of Area): 69,00% Estricção
Alongamento (Elongation): 42,00% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 12,70 mm (1/2") (BR 19/01422/1)-(34 pcs)

Num. Rast. BR 19/01422/1

Corrida (Heat): 111236

Placa/O.P.:

Norma (Standard): ASTM A276/M-17 A

Peso (Weight): 102,00 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Analise Quimica (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,023	0,470	1,6	18,12	8,07	0,4	0,0750		0,037
S	Cu	Al	N2	Nb	Co			
0,027	0,59				0,14			

Analise Mecanica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 571 MPa
Resistencia a Tracao (Strizione): 699 MPa Dureza: 207 HBW
Reduc. Area Minima (Reduction of Area): Estricção 69,00%
Alongamento (Elongation): 38,00% Tam. do Grao (Grain Size):



Aperam Inox América do Sul S.A.

aperam

Sede: Av. Carandá, nº 1115, 23º Andar - Centro - Belo Horizonte - MG - 30130-915 - Brasil
 Usina: Praça 1º de Maio, nº 9 - Centro - Timóteo - MG - 35180-018 - Brasil
 Filiais: Av. Mercedes Benz, nº 1.420 - Distrito Industrial - Campinas - SP - 13054-750 - Brasil
 Rodovia RSC 453, Km 2,2 - S/N - Distrito Industrial - Caxias do Sul - RS - 95110-690 - Brasil

CERTIFICADO DE QUALIDADE

Número: 2178455
 Pedido APERAM: 82875034
 Data: 29/07/2019
 Nota Fiscal: 838063

Cliente:
 JATI-SERVICOS COMERCIO IMP. ACOS
 RUA ROBERTO KOCH 363
 04221060 - SAO PAULO - SP
 BRASIL
 Ordem: 11058

Aço: AISI 420 Acabamento: 2B Produto: BOBINA
 Dimensões: 1,50 mm x 1240 mm
 Marcação:

Composição Química

Número Corrida	C %	Mn %	Si %	P %	S %	Cr %	Ni %	Mo %	N PPM	Ti %	Cu %	Co %				
970914H	0,313	0,32	0,370	0,0251	0,0032	12,07	0,15	0,012	112	0,0029	0,0289	0,015				

Resultados

Ensaio/Variável	Unid. Med	970914H2003B													
		Inicial	Final	Inicial	Final	Inicial	Final	Inicial	Final	Inicial	Final	Inicial	Final	Inicial	Final
Dureza RB	HRB	85,33	85,33												
Peso Líquido	Kg	9.070													
Peso Bruto	Kg	9.080													

Identificação de unidade metálica (U.M.):

Ok

Qualidade superficial / dimensional:

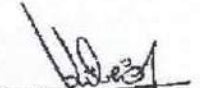
Ok

Normas: ASTM A240/A240M 18

ASTM A176 _ 99 04

Observações: LARGURAS REAIS DE REFERÊNCIA:
 970914H2003B: Mín. 1252 mm - Máx. 1252 mm

- a) Material sem contaminação por Mercúrio.
- b) Material sem contaminação por radioatividade. (Garantia enquanto estocado sob responsabilidade da Aperam Inox América do Sul S.A.).
- c) Certificamos que os resultados descritos acima estão corretos e de acordo com a especificação ou concessão autorizada.
- d) Sistema de Gestão da Qualidade certificado conforme norma ISO 9001 - Certificado ABS QE Nº 30071.
- e) Sistema de Gestão da Qualidade certificado conforme norma ISO/TS 16949 - Certificado ABS QE Nº 38324.
- f) Sistema de Gestão Ambiental certificado conforme norma ISO 14001 - Certificado ABS QE Nº 65989.
- g) Corrida(s) produzida(s) no Brasil.


 Paulo Sérgio Ribeiro
 Inspetor Qualificado
 Aperam Inox América do Sul S.A.



TEST CERTIFICATE

CUSTOMER TROP COMERCIO EXTERIOR Ltda. AV. JERONIMO MONTEIRO, 1000-SALAS 1718, 1720, 1722 e 1724. EDIFICIO TRADE CENTER CENTRO-VITORIA-ES BRASIL CEP BUNDLE NO : 5186033 , 5186030 , 5186038 , 5186039 ,	ORDER NO SC-02/11	PACKING LIST NO 51/936152/1	INSPECTION NO 100001566318 GRADE AISI 420	DATE 01.03.2012 HEAT No 25056
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DESCRIPTION

STAINLESS STEEL BRIGHT BARS

COLD DRAWN POLISHED - 2H

SIZE	SHAPE	TOLERANCE	LENGTH	PIECES	WEIGHT
1/4"	ROUND	h9	5.00 - 5.20	2945	3.694 TO

CHEMICAL ANALYSIS

%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N2
0.1500	0.760	0.460	0.0100	0.0210	12.140	0.290	0.080	0.030	0.050	0.00400

TEST RESULTS

0.2% yield strength MPa	Tensile Strength MPa	Elongation %	Reduction Of Area %	Hardness BHN
631.0	718.0	25.0	65.0	213.0

SPECIFICATION:

MATERIAL CONFIRMS TO ASTM A 276-10 COND A.

CERTIFIED AS PER EN 10204-3.1

(HEAT TREATMENT # ANNEALED AT 865°C (±5° C) SOAKING 8 HRS (+ - 1 HRS) FURNACE COOLED)

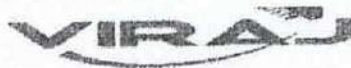
**REMARKS**

MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS

WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION

We hereby certify that the material described above has been tested and complies with the terms of order/contract.

NR
BR12-263

**VIRAJ PROFILES LIMITED (SRM Division)**

Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHTRA -401 506, INDIA.
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's	(A03)MTR NO.	1056011/4
	Symbol (A04)	(Z02) DATE	20.09.2019
		MATERIAL SPEC.	
		(B02) GRADE	AISI 304L
		DIMENSIONAL SPEC.	h9
(B01) : STAINLESS STEEL BRIGHT BARS		(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H	

(A07)ORDER NO: JATI013-19			(A09)Article No:	
(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 5/16"	5.00 - 6.00	845	1.950	112704
Bundle No	1101160954,1101160955			

(C71-C92) CHEMICAL ANALYSIS

ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo		%Co	%N			
HEAT	0.0230	1.5300	0.5300	0.0240	0.0370	18.1500	8.0500	0.7000	0.3700		0.1500	0.0720			

MECHANICAL PROPERTIES

Inspection No.														
100011832833	(C12)	Yield Strength		(C13)	Reduction of Area%		Hardness							
	Tensile Strength	Rp=0.2%	Rp = 1%	Elongation%			HBW							
	Mpa	Mpa	Mpa	Lo=4D	71.00		228							
	772.00	650.00		37.00										

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.



REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C, SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY

**VIRAJ PROFILES LIMITED (SRM Division)**

Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHTRA -401 506, INDIA,
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL. CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's	(A03)MTR NO.	1052245/3
	Symbol (A04)	(Z02) DATE	09.06.2019
		MATERIAL SPEC.	
		(B02) GRADE	AISI 304L
		DIMENSIONAL SPEC.	h9

(B01) : STAINLESS STEEL BRIGHT BARS**(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H****(A07)ORDER NO: JAT1007-19****(A09)Article No:**

(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 1 1/8"	5.00 - 6.00	39	1.196	53335

Bundle No 1101105484,1101105485

(C71-C92) CHEMICAL ANALYSIS

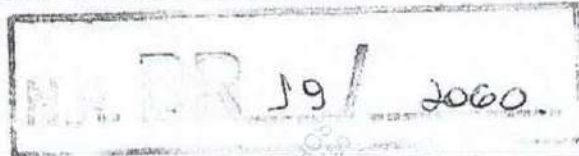
ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N				
HEAT	0.0160	1.5500	0.3500	0.0220	0.0370	18.2000	8.0800	0.5700	0.3400	0.1600	0.0700				

MECHANICAL PROPERTIES

Inspection No.															
100011477383	(C12)	Yield Strength		(C13)		Reduction of Area%		Hardness							
	Tensile Strength	Rp=0.2%	Rp = 1%	Elongation%											
	Mpa	Mpa	Mpa	Lo=4D				HBW							
	675.00	494.00		42.00			69.00	200							

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.



REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C, SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY



**VIRAJ PROFILES LIMITED (SRM Division)**

Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHITRA -401 506, INDIA,
e-mail- viraj@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's Symbol (A04) 	(A03)MTR NO.	1050091/3
		(Z02) DATE	04.04.2019
		MATERIAL SPEC.	
		(B02) GRADE	AISI 304L
		DIMENSIONAL SPEC.	h9

(B01) : STAINLESS STEEL BRIGHT BARS (B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H

(A07)ORDER NO: JAT1005-19			(A09)Article No:	
(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 1/2"	5.00 - 6.00	2258	13.304	111236

Bundle No 1101071364,1101071366,1101071370,1101071362,1101071360,1101071357,1101071372,1101071371,1101071359,1101071363,1101071365,1101071369,1101071356,1101071361

(C71-C92) CHEMICAL ANALYSIS

ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N				
HEAT	0.0230	1.6000	0.4700	0.0270	0.0370	18.1200	8.0700	0.5900	0.4000	0.1400	0.0750				

MECHANICAL PROPERTIES

Inspection No. 100011258176														
	(C12) Tensile Strength	Yield Strength		(C13) Elongation%		Reduction of Area%		Hardness						
		Rp=0.2%	Rp = 1%											
		Mpa	Mpa	Lo=4D				HBW						
		699.00	571.00	38.00		69.00		207						

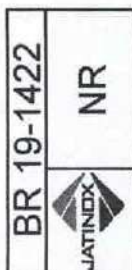
(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A.
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.

19 1422

REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C. SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



28 JUL 2020




AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY



Certificado de Qualidade Quality Certificate



785576/2020

Cliente (Customer): MACOM INSTRUMENTAL CIRURGICO IND LTDA

Pedido Cliente (Customer order): 13160

Pedido(Order): 762543

Nota Fiscal (Invoice): 388836

Data (Date): 22/04/2020

Material (Product)

Descricao (Description): Barra Redonda 304L - 7,94 mm (5/16") (BR 20/00675/1)-(10 pcs)

Num. Rast. BR 20/00675/1

Corrida (Heat): 113612

Placa/O.P.:

Norma (Standard): ASTM A276/A276M

Peso (Weight): 23,70 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,021	0,260	1,55	18,15	8,06	0,33	0,075		0,035
S	Cu	Al	N2	Nb	Co			
0,025	0,38				0,15			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 632 MPa
Resistencia a Tracao (Strizione): 758 MPa Dureza: 225 HBW
Reduc. Area Minima (Reduction of Area): 70 Estricção
Alongamento (Elongation): 42 Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 14,29 mm (9/16") (BR 20/00622/1)-(5 pcs)

Num. Rast. BR 20/00622/1

Corrida (Heat): 0813784

Placa/O.P.:

Norma (Standard): ASTM A276/A276M

Peso (Weight): 38,00 KG

Observacao (Observation): BARRA REDONDA / LAMINADOS / SOLUBILIZADO NA LAMINAÇÃO / DESCASCADO POLIDO

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,026	0,230	1,5	18,32	8,33	0,63	0,095		0,026
S	Cu	Al	N2	Nb	Co	V	W	
0,027	0,62				0,13	0,069	0,11	

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 259 MPa
Resistencia a Tracao (Strizione): 581 MPa Dureza: 151,4 HBW
Reduc. Area Minima (Reduction of Area): Estricção 78,0%
Alongamento (Elongation): 59,4% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 420 - 7,94 mm (5/16") (BR 17/00293/1)-(15 pcs)

Num. Rast. BR 17/00293/1

Corrida (Heat): 43246

Placa/O.P.:

Norma (Standard): ASTM A276/A276M

Peso (Weight): 32,50 KG

Observacao (Observation):

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,300	0,410	0,52	12,12	0,42	0,05			0,031
S	Cu	Al	N2	Nb	CO			
0,012			0,02900		0,03			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 696,0 MPa
Resistencia a Tracao (Strizione): 799,0 MPa Dureza: 237,0 BHN
Reduc. Area Minima (Reduction of Area): 58,0% Estricção
Alongamento (Elongation): 20,0% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 420 - 6,35 mm (1/4") (BR 08/00328/1)-(20 pcs)

Num. Rast. BR 08/00328/1

Corrida (Heat): 12320262

Placa/O.P.:

Norma (Standard):

Peso (Weight): 24,80 KG

Observacao (Observation):

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,370	0,340	0,37	13,02	0,52	0,09	0,0170	0	0,038
S	Cu	Al	N2	Nb	Co			
0,017	0,09	0	0	0	0	0	0	0

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 790 MPa
Resistencia a Tracao (Strizione): 851 MPa Dureza: 260 HB
Reduc. Area Minima (Reduction of Area): Estricção 45
Alongamento (Elongation): 14 Tam. do Grao (Grain Size):





Certificado de Qualidade Quality Certificate



785576/2020

Cliente (Customer): MACOM INSTRUMENTAL CIRURGICO IND LTDA

Nota Fiscal (Invoice): 388836

Pedido Cliente (Customer order): 13160

Pedido(Order): 762543

Data (Date): 22/04/2020

Material (Product)

Descricao (Description): Barra Redonda 420 - 25,40 mm (1") (BR 19/02101/1)-(5 pcs)

Num. Rast. BR 19/02101/1

Corrida (Heat): 0812965

Placa/O.P.:

Norma (Standard): ASTM A276/17

Peso (Weight): 115,00 KG

Observacao (Observation): BARRA REDONDA / LAMINADOS / RECOZIDO / DESCASCADO POLIDO

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,034	0,420	0,4	12,4	0,3				0,029
S	Cu	Al	N2	Nb		B		
0,026	0,03			0,03		0,0005		

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 417 MPa
Resistencia a Tracao (Strizione): 679 MPa Dureza: 204 HBW
Reduc. Area Minima (Reduction of Area): Estricção 51,4%
Alongamento (Elongation): 28,1% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 420 - 15,87 mm (5/8") (BR 18/01231/1)-(5 pcs)

Num. Rast. BR 18/01231/1

Corrida (Heat): 49595

Placa/O.P.:

Norma (Standard): ASTM A276M-17 A

Peso (Weight): 47,00 KG

Observacao (Observation):

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,170	0,450	0,75	12,06	0,18	0,02			0,025
S	Cu	Al	N2	Nb	Co			
0,026	0,06		0,0340	0,03				

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 605 MPa
Resistencia a Tracao (Strizione): 720 MPa Dureza: 214 HBW
Reduc. Area Minima (Reduction of Area): 65,00% Estricção
Alongamento (Elongation): 21,00% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 9,53 mm (3/8") (BR 19/02530/1)-(10 pcs)

Num. Rast. BR 19/02530/1

Corrida (Heat): 0813010

Placa/O.P.:

Norma (Standard): ASTM A276/17

Peso (Weight): 27,80 KG

Observacao (Observation): BARRA REDONDA / LAMINADOS / SOLUBILIZADO NA LAMINAÇÃO / TREFILADO RETIFICADO

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,024	0,200	1,54	18,14	8,34	0,62	0,097		0,031
S	Cu	Al	N2	Nb	Co	V	W	
0,027	0,62				0,22	0,076	0,11	

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 459 MPa
Resistencia a Tracao (Strizione): 692 MPa Dureza: 274,5 HBW
Reduc. Area Minima (Reduction of Area): Estricção 73,9%
Alongamento (Elongation): 50,0% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 6,35 mm (1/4") (BR 20/00612/1)-(17 pcs)

Num. Rast. BR 20/00612/1

Corrida (Heat): 113509

Placa/O.P.:

Norma (Standard): ASTM A276/A276M

Peso (Weight): 24,70 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,027	0,230	1,66	18,2	8,1	0,4	0,075		0,037
S	Cu	Al	N2	Nb	Co			
0,016	0,55				0,16			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 693 MPa
Resistencia a Tracao (Strizione): 796 MPa Dureza: 236 HBW
Reduc. Area Minima (Reduction of Area): Estricção 100,00%
Alongamento (Elongation): 35,00% Tam. do Grao (Grain Size):





Certificado de Qualidade Quality Certificate



785576/2020

Cliente (Customer): MACOM INSTRUMENTAL CIRURGICO IND LTDA

Pedido Cliente (Customer order): 13160

Pedido(Order): 762543

Nota Fiscal (Invoice): 388836

Data (Date): 22/04/2020

Material (Product)

Descricao (Description): Barra Redonda 304L - 19,05 mm (3/4") (BR 19/01785/1)-(10 pcs)

Num. Rast. BR 19/01785/1

Corrida (Heat): 53616

Placa/O.P.:

Norma (Standard): ASTM A276/M-17 A Peso (Weight): 137,00 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,014	0,250	1,56	18,2	8,06	0,3	0,0810		0,036
S	Cu	Al	N2	Nb	Co			
0,025	0,49				0,17			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 613 MPa
Resistencia a Tracao (Strizione): 742 MPa Dureza: 220 HBW
Reduc. Area Minima (Reduction of Area): 67,00% Estricção
Alongamento (Elongation): 36,00% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 15,87 mm (5/8") (BR 20/00539/1)-(10 pcs)

Num. Rast. BR 20/00539/1

Corrida (Heat): 56263

Placa/O.P.:

Norma (Standard): ASTM A276/A276M Peso (Weight): 95,00 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,012	0,320	1,56	18,2	8,1	0,39	0,0760		0,035
S	Cu	Al	N2	Nb	Co			
0,023	0,54				0,16			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 636 MPa
Resistencia a Tracao (Strizione): 750 MPa Dureza: 222 HBW
Reduc. Area Minima (Reduction of Area): 69,00% Estricção
Alongamento (Elongation): 38,00% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 304L - 12,70 mm (1/2") (BR 20/00560/1)-(20 pcs)

Num. Rast. BR 20/00560/1

Corrida (Heat): 56267

Placa/O.P.:

Norma (Standard): ASTM A276/A276M Peso (Weight): 121,00 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 1050°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,014	0,250	1,5	18,25	8,07	0,31	0,0810		0,034
S	Cu	Al	N2	Nb	Co			
0,02	0,59				0,15			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 668 MPa
Resistencia a Tracao (Strizione): 776 MPa Dureza: 230 HBW
Reduc. Area Minima (Reduction of Area): 70,00% Estricção
Alongamento (Elongation): 34,00% Tam. do Grao (Grain Size):

Material (Product)

Descricao (Description): Barra Redonda 420 - 14,29 mm (9/16") (BR 19/02796/1)-(3 pcs)

Num. Rast. BR 19/02796/1

Corrida (Heat): 112253

Placa/O.P.:

Norma (Standard): ASTM A276/M-17 A Peso (Weight): 21,80 KG

Observacao (Observation): TREFILADO A FRIO / POLIDO / RECOZIDO 865°C

Análise Química (Chemical Composition)

C	SI	Mn	Cr	Ni	Mo	N	Ti	P
0,0170	0,330	0,72	12,14	0,49	0,08	0,0390		0,029
S	Cu	Al	N2	Nb	Co			
0,02	0,12				0,02			

Análise Mecânica (Mechanical Properties)

Limite de Escoamento (Yield Stress): 0,2% 652 MPa
Resistencia a Tracao (Strizione): 736 MPa Dureza: 218 HBW
Reduc. Area Minima (Reduction of Area): 62,00% Estricção
Alongamento (Elongation): 22,00% Tam. do Grao (Grain Size):





VIRAJ PROFILES LIMITED (SRM Division)
Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHTRA -401 506, INDIA,
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's Symbol (A04)	(A03)MTR NO.	1060115/8
		(Z02) DATE	10.01.2020
		MATERIAL SPEC.	
		(B02) GRADE	AISI 304L
		DIMENSIONAL SPEC.	h9
(B01) : STAINLESS STEEL BRIGHT		(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H	

(A07)ORDER NO: JATI025-19		(A09)Article No:		
(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 1/2"	5.00 - 6.00	651	3.872	56267
Bundle No	1101218368,1101218364,1101218369,1101218365			

(C71-C92) CHEMICAL ANALYSIS

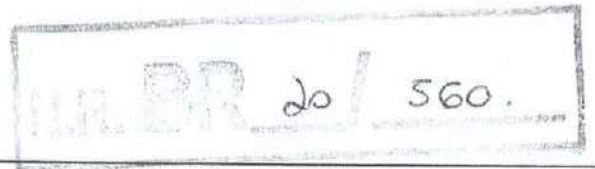
ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N				
HEAT	0.0140	1.5000	0.2500	0.0200	0.0340	18.2500	8.0700	0.5900	0.3100	0.1500	0.0810				

MECHANICAL PROPERTIES

Inspection No.										
100012206638	(C12) Tensile Strength	Ycild Strength		(C13) Elongation%	Reduction of Area%	Hardness				
		Rp=0.2%	Rp = 1%							
		Mpa	Mpa	Mpa	Lo=4D		HBW			
		776.00	668.00		34.00	70.00	230			

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.



REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C, SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY



VIRAJ PROFILES LIMITED (SRM Division)
Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHTRA -401 506, INDIA.
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL. CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's Symbol (A04) 	(A03)MTR NO.	1055668/8
		(Z02) DATE	10.09.2019
		MATERIAL SPEC.	
		(B02) GRADE	AISI 420
		DIMENSIONAL SPEC.	h9

(B01) : STAINLESS STEEL BRIGHT BARS,ANGLE (B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H

(A07)ORDER NO: JATI011-19

(A09)Article No:

(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 9/16"	5.00 - 6.00	149	1.148	112253

Bundle No 1101155624,1101155623

(C71-C92) CHEMICAL ANALYSIS

ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N				
HEAT	0.1700	0.7200	0.3300	0.0200	0.0290	12.1400	0.4900	0.1200	0.0800	0.0200	0.0390				

MECHANICAL PROPERTIES

Inspection No.															
100011799410	(C12) Tensile Strength	Yield Strength		(C13) Elongation%		Reduction of Area%		Hardness							
		Rp=0.2%	Rp = 1%												
	Mpa	Mpa	Mpa	Lu=4D				HBW							
	736.00	652.00		22.00		62.00		218							

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
CERTIFIED AS PER EN 10204-3.1.
(HEAT TREATMENT # ANNEALED AT 865°C (±5° C) SOAKING 8 HRS { +/- 1HRS} FURNACE COOLED).



REMARKS :

MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin

TABELÃO DE NOTAS DE GARANTIA
Mauro Alexandre Barbosa
Av. Dr. Tereza Pontes, 0071 - VI. Heliópolis
AUTENTICAÇÃO



30 ABR 2020

AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY





Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHITRA -401 506, INDIA.
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06) CUSTOMER :

COMEXPORT TRADING COMERCIO

RUA JOAO BAUER NO 498

SALAS 802 803 E 804

CENTRO-ITAJAI-SC-BRASIL

CEP 88301-500 CNPJ: 01.135.153/0003-70

Manufacturer's

Symbol (A04)



(A03) MTR NO.

1051409/3

(Z02) DATE

18.05.2019

MATERIAL SPEC.

(B02) GRADE

ASTI 304L

DIMENSIONAL SPEC.

h9

(B01) : STAINLESS STEEL BRIGHT BARS

(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H

(A07) ORDER NO: JATI005-19-COMP

(B09-B11) ITEM DESCRIPTION

LENGTH

(B08) PIECES(Nos)

(A09) Article No:

ROUND 3/4"

5.00 - 6.00

NET WEIGHT(TO)

(B07) HEAT NO

68

0.916

53616

Bundle No

1101091740

(C71-C92) CHEMICAL ANALYSIS

ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N
HEAT	0.0140	1.5600	0.2500	0.0250	0.0360	18.2000	8.0690	0.4900	0.3000	0.1700	0.0810

MECHANICAL PROPERTIES

Inspection No.	(C12) Tensile Strength	Yield Strength	(C13) Elongation%	Reduction of Area%	Hardness
100011396793	Rp=0.2%	Rp = 1%	Le=4D		HBW
	Mpa	Mpa	Mpa		
	742.00	613.00	36.00	67.00	220

(B06) Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A.
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.

19 1785.

REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C, SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY



VILLARES METALS

CERTIFICADO DE QUALIDADE

CONFORME EN 10204 3.1

CERTIFICADO 922191

Pag. 1/ 1

NOTA FISCAL 255341

Nosso sistema de qualidade é certificado de acordo com ISO 9001, IATF 16949, NBR 15100 (AS 9100) e NADCAP (tratamento térmico e ultrassom).
Certificação Ambiental ISO 14001.

CLIENTE **JATI SERVICOS COMERCIO E IMPORTACAO DE ACOS LTDA**

PEDIDO DO CLIENTE / SEQUÊNCIA / ITEM **JATI 008-19 C- / 27.06.2019**

OP	LOTE	VOLUME	PESO	QUANTIDADE
1815002	1815002	1815002/01	2.126 KG	270
LIGA VILLARES		LIGA CLIENTE/SIMILAR		
V304LT		AISI 304L/S30403		
DIMENSÕES		COMPRIMENTO (mm)	Nº DO DESENHO	
9,53 Tol (-)0,036 (+)0,000		4.500 / 5.000		
PRODUTO		TRATAMENTO TÉRMICO	ACABAMENTO	
Barra Redonda		Solubilizado na Laminação	Trefilado Retificado	
LINHA DE PRODUTO		PROCESSO DE FUSÃO	CORRIDA	
Laminados		FEA (LING. CONTÍNUO)	0813010	
COMPOSIÇÃO QUÍMICA (% em peso)				
Análise química na liga				
C 0,024	Si 0,20	Mn 1,54	P 0,031	S 0,027
Co 0,22	Cr 18,14	Mo 0,62	Ni 8,34	V 0,076
W 0,11	Cu 0,62	N 0,097		
NORMAS ATENDIDAS				
ASTM A276 / 17				
Tamanho de grão conforme norma ASTM E112				
Situação: Aprovado				
TAMANHO DE GRÃO ENCONTRADO: 8 ASTM				
Tração - longitudinal conforme norma ASTM A370				
Corpo de prova n°: 43586		Posição: Longitudinal	Temperatura: 23,90 °C	
Dimensão: 10,18 mm		Lim. resist.: 692 MPa	Lim. esc. 0,2%: 459 MPa	
Estricção: 73,9 %		Alongamento: 50,0 %		
Dureza conforme norma ASTM E10				
Resultado: 274,5 HBW				
Corrosão conforme norma ASTM A262 Met.A				
Situação: Aprovado				
Ultrassom conforme norma ASTM A388				
Situação: Aprovado				
Inspeção visual e dimensional				
Situação: Aprovado				
Teste anti-mistura				
Situação: Aprovado				
A VILLARES METALS DECLARA QUE ESTE PRODUTO ESTÁ EM CONFORMIDADE COM AS ESPECIFICAÇÕES DO PEDIDO DE COMPRA				
USINA DE SUMARÉ / ESCRITÓRIO CENTRAL R. Alfredo Dumont Villares, 155 13178-902 - Sumaré - SP - BRAZIL Fone: 55 19 3303-8000 Fax: 55 19 3303-8696 e-mail: metals@villaresmetals.com.br			08.10.2019 José Roberto Moreira Melo Gerente de Engenharia de Produto	



NR. BR 19 / 2530.

TABELÃO DE NOTAS DE GUARULHO
Mauro Alexandre Barbosa Bordini - Tabelão
Autenticado
08.10.2019

BR 19-2530

NR





VIRAJ PROFILES LIMITED (SRM Division)
Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- PALGHAR, MAHARASHTRA -401 506, INDIA,
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : COMEXPORT TRADING COMERCIO RUA JOAO BAUER NO 498 SALAS 802 803 E 804 CENTRO-ITAJAI-SC-BRASIL CEP 88301-500 CNPJ: 01.135.153/0003-70	Manufacturer's Symbol (A04)	(A03)MTR NO. 1060084/7
		(Z02) DATE 09.01.2020
		MATERIAL SPEC.
		(B02) GRADE AISI 304L
		DIMENSIONAL SPEC. h9

(B01) : STAINLESS STEEL BRIGHT BARS	(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H
-------------------------------------	---

(A07)ORDER NO: JATI022-19-A	(A09)Article No:
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(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 5/8"	5.00 - 6.00	294	2.752	56263

Bundle No	1101217906,1101217905,1101217904
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(C71-C92) CHEMICAL ANALYSIS

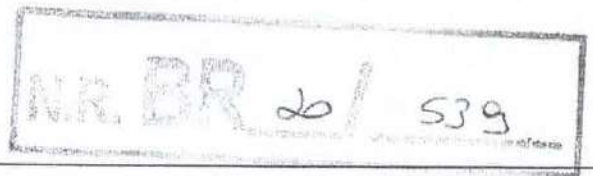
ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N
HEAT	0.0120	1.5600	0.3200	0.0230	0.0350	18.2000	8.1000	0.5400	0.3900	0.1600	0.0760

MECHANICAL PROPERTIES

Inspection No.	(C12) Tensile Strength	Yield Strength	(C13) Elongation%	Reduction of Area%	Hardness
		Rp=0.2% Rp = 1%			
	Mpa	Mpa	Lo=4D		HBW
100012203351	750.00	636.00	38.00	69.00	222

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
INTERGRANULAR CORROSION TEST SATISFACTORY AS PER ASTM A262-15 PRACTICE E.
CERTIFIED AS PER EN 10204-3.1.



REMARKS : MINIMUM SOLUTION ANNEALING TEMPERATURE 1050°C. SOAKING TIME 01 HR/INCH WATER QUENCHED.
MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.
WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin



AUTHORIZED WORKS INSPECTOR
RADHAKRISHNA MURTHY

**VIRAJ PROFILES LIMITED (SRM Division)**

Works: SURVEY NO.- 25/1, 25/2, 26 & 34, Village- MAAN, Taluka- PALGHAR, Dist.- THANE, MAHARASHTRA -401 506, INDIA.
e-mail- vielqc@viraj.com

(A02) INSPECTION CERTIFICATE & MILL TEST REPORT

(A06)CUSTOMER : JATI SERVICOS COMERCIO E IMPORTACAO RUA ROBERTO KOCH,363 Ipiranga - SAO PAULO/SP CEP: 04221-060 / CNPJ: 03.846.642/0001-0	Manufacturer's	(A03)MTR NO.	1038594/17
	Symbol (A04)	(Z02) DATE	24.05.2018
		MATERIAL SPEC.	
		(B02) GRADE	AISI 420
		DIMENSIONAL SPEC.	H9

(B01) : STAINLESS STEEL BRIGHT BARS**(B04) DELIVERY CONDITION : COLD DRAWN POLISHED - 2H****(A07)ORDER NO: JATI2202****(A09)Article No:**

(B09-B11)ITEM DESCRIPTION	LENGTH	(B08) PIECES(Nos)	NET WEIGHT(TO)	(B07) HEAT NO
ROUND 5/8"	5.00 - 6.00	102	0.950	49595

Bundle No 1100901344

(C71-C92) CHEMICAL ANALYSIS

ELEMENT	%C	%Mn	%Si	%S	%P	%Cr	%Ni	%Cu	%Mo	%Co	%N2				
HEAT	0.1700	0.7600	0.4500	0.0260	0.0250	12.0600	0.1800	0.0600	0.0200	0.0300	0.0340				

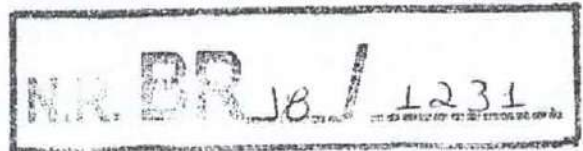
MECHANICAL PROPERTIES

Inspection No.												
	(C12) Tensile Strength	Yield Strength		(C13) Elongation%	Reduction of Area%		Hardness					
100010157810	Mpa	Rp=0.2%	Rp = 1%	Lo=5D			HBW					
	720.00	605.00		21.00	65.00		214					

(B06)Product Marking : SIZE, GRADE, HEAT NO., LOGO & INSPECTOR STAMP

MATERIAL CONFIRMS TO ASTM A276/A276M-17 COND A,
CERTIFIED AS PER EN 10204-3.1.

(HEAT TREATMENT # ANNEALED AT 865°C (±5° C) SOAKING 8 HRS { +/- 1HRS} FURNACE COOLED).

**REMARKS :**

MATERIAL IS FREE FROM MERCURY CONTAMINATION, FREE FROM WELD OR WELD REPAIRS.

WE CONFIRM THAT THE MATERIAL HAS BEEN TESTED AND FOUND TO BE FREE FROM RADIOACTIVE CONTAMINATION.

we certify that the above material has been inspected and tested and complies with the order/contract and is of Indian origin.

**AUTHORIZED WORKS INSPECTOR****RADHAKRISHNA MURTHY****NR
BR18-1231**

JATI-SERVIÇOS COMÉRCIO E IMPORTAÇÃO DE AÇOS LTDA



JATINOX
JATI-SERVIÇOS COM. E IMP. DE AÇOS LTDA.



ISO 9001:2000

Certificado desde 1999



Rua Roberto Koch, 363 - Cep 04221-060 - Ipiranga - São Paulo - SP - Site : www.jati.com.br - E-mail : vendas@jati.com.br
Fone Tronco: (011) 2172-0405 Fax Cobrança : (011) 2172-0432
CNPJ : 03.846.642/0001-02 Inscr. Est. 115.719.976.110

DECLARAÇÃO

A **JATI-SERVIÇOS COMÉRCIO E IMPORTAÇÃO DE AÇOS LTDA**, inscrita no CNPJ sob o nº 03.846.642/0001-02, com sede à Rua Roberto Koch, nº 363 – Bairro Ipiranga, cidade de São Paulo, estado de São Paulo, declara por meio desta, ser fornecedor regular de Aço Inoxidável, da empresa **MACOM INSTRUMENTAL CIRURGICO INDUSTRIA LTDA**, inscrita no CNPJ sob o nº 59.650.556/0001-76. Informamos ainda que o aço inoxidável fornecido atende aos requisitos da norma ABNT NBR ISO 7153-1 : 1997.

São Paulo, 15 de Setembro de 2.010

Delorde de Oliveira Mota

DELORDE DE OLIVEIRA MOTA
CPF : 620.436.688-20

